

Sanjay Keerthi

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PROFESSIONAL SUMMARY

Data Scientist experienced in cleaning, validating, and modelling large, messy real-world datasets, with a track record of turning analysis into dashboards and reporting that speeds up decision-making. Builds automated data pipelines, runs exploratory and statistical analysis, and works across deep learning, NLP, and time-series forecasting to solve real business problems. MSc Data Science candidate at the University of Manchester, focused on Reinforcement Learning, Bayesian Inference, and probabilistic modelling.

Combines technical depth with strong business understanding: hands-on retail and hospitality experience managing stock control, food waste reduction, and front-to-back operations at Grand Kerala gives practical understanding of how goods move through a supply chain and where costs build up. Skilled at root causing issues, spotting data quality problems, and turning findings into clear insight for stakeholders. Comfortable leading own projects and delivering real change to how things work.

TECHNICAL SKILLS

Languages: Python, SQL, R

ML / AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Keras, Hugging Face, Ollama

ML Techniques: Supervised & Unsupervised Learning, Deep Learning, NLP, Computer Vision, Time-Series Forecasting, Reinforcement Learning, Bayesian Inference, Dimensionality Reduction (PCA, t-SNE), Hyperparameter Optimisation, Overfitting Detection, Data Leakage Identification, Model Deployment

Business & Analytics: Root Cause Analysis, Demand Forecasting, Cost & Waste Optimisation, Stakeholder Communication, Data-Driven Decision Making

Data & Cloud: Pandas, NumPy, SciPy, Statsmodels, AWS, Snowflake, MLflow, Airflow

Practices: Feature Engineering, Cross-Validation Strategies, Imbalanced Dataset Handling, AI Evaluation, Data Privacy & Security, Technical Documentation, Git, Docker

PROFESSIONAL EXPERIENCE

Data Scientist Intern — Vcodez (Remote)

Jan 2025 – Jul 2025

- Built automated data ingestion, cleaning, and quality-validation pipelines in Airflow for 10M+ row climate-monitoring datasets, identifying and resolving inconsistencies before analysis and cutting 6+ hours of manual processing per sprint; documentation adopted by the engineering team.
- Validated PyTorch and TensorFlow model outputs before they informed decisions, auditing for overfitting and data-leakage issues and turning findings into technical feedback that improved production model performance.
- Engineered LSTM and ARIMA models on the validated pipelines to support real-time forecasting, achieving 91% accuracy and enabling data-driven decisions for business partners across the network.

Data Analytics Intern — Preplnsta (Remote)

Jun 2024 – Oct 2024

- Performed exploratory data analysis and statistical analysis on customer campaign-spend data to identify data quality issues and inform feature engineering; partnered with a 4-member team on cross-validation strategy, delivering a 15% F1-score improvement on imbalanced classification models.
- Cut reporting turnaround by 25% using Python-based EDA, data mining, and statistical analysis, improving how quickly stakeholders received insight into performance changes.

PROJECTS

Domain-Robust Polyp Detection — LightPolyp (MSc Dissertation)

Python, PyTorch

- Working with clinical colonoscopy imaging datasets from multiple sources; addressed cross-domain data inconsistency in LightPolyp, a lightweight (~220k-parameter) architecture with a style-normalisation head, depthwise separable backbone, and case-level splitting to prevent data leakage, benchmarking generalisation across Kvasir-SEG, CVC-ClinicDB, and ETIS-LaribPolypDB to understand where and why performance changes across clinical datasets.

Sports Analytics Dashboards — FIFA & IPL

Python, Streamlit, XGBoost

- Built interactive dashboards (Streamlit) to make fragmented public sports data (FIFA Club World Cup, IPL) accessible for tracking performance, with custom venue-normalisation utilities delivering business-ready insight from raw data; also engineered direct FIFA API scraping and an ELO-based prediction system (XGBoost, Poisson, Dixon-Coles) across five targets.

EDUCATION

MSc Data Science — The University of Manchester | Sep 2025 – Sep 2026

Core modules: Statistics & Machine Learning, Reinforcement Learning, Data Engineering Technologies

B.E. Computer Science (CGPA 8.58/10) — Panimalar Engineering College | Aug 2021 – May 2025

CERTIFICATIONS

ML & Deep Learning Specialisation, Coursera/DeepLearning.AI (2024) | GitHub Foundations (2024–2027) | Oracle Certified Foundation Associate (2024–2026)